

Software Requirements Specification (SRS)

Municipal Solid Waste Route Optimizer

Standard IEEE 830-1998 Specification Format | Student SRN: PES1UG24CS453

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Municipal Solid Waste Route Optimizer System. The system ingests ultrasonic fill-level telemetry from smart trash receptacles, calculates dynamically optimized collection routes for city garbage trucks under stringent real-time constraints, and provides full lifecycle tracking for missed-pickup citizen complaints.

1.2 Scope

The Municipal Solid Waste Route Optimizer replaces static garbage pickup schedules with real-time, demand-driven dispatching. It monitors 10,000 smart trash bins equipped with ultrasonic sensors, dynamically includes only receptacles over 80% full, recalculates coordinated routes for 50 collection trucks in under 45 seconds, delivers digital manifests to truck drivers, and provides citizen complaint registration.

1.3 Definitions, Acronyms, and Abbreviations

- VRP: Vehicle Routing Problem (combinatorial NP-hard optimization).
- IoT: Internet of Things (smart ultrasonic bin sensors).
- RBAC: Role-Based Access Control.
- SLA: Service Level Agreement (sub-45s route calculation constraint).
- MQTT: Message Queuing Telemetry Transport (lightweight sensor messaging protocol).
- gRPC: High-performance, open-source universal RPC framework.

2. Overall Description

2.1 Product Perspective

The system operates as a cloud-native microservices architecture comprising an API Gateway, Sensor Data Ingestion Service, Route Optimization Service, Driver Manifest Service, Complaint Management Service, and a multi-model spatial/time-series persistence tier.

2.2 User Classes and Characteristics

1. Sanitation Supervisor: Operations manager overseeing city-wide waste collection, monitoring telemetry graphs, initiating route recalculation, and inspecting complaint resolution.
2. Truck Driver: Field operator navigating in-cab navigation terminals to follow assigned collection manifests and acknowledge pickups.
3. Citizen: Public user reporting missed garbage collections or overflow incidents via citizen portal.
4. System Administrator: IT personnel maintaining gateway configurations, security certificates, and service scaling.

2.3 Operating Environment & Constraints

The system runs on containerized Linux nodes (Kubernetes) backed by PostgreSQL with PostGIS spatial extensions, TimescaleDB time-series storage, and Redis cache. Truck terminals operate on Android/iOS via progressive web application / native mobile frameworks.

3. Specific Functional Requirements

FR-1: Ultrasonic Telemetry Ingestion: The system shall continuously ingest ultrasonic fill-level readings from up to 10,000 smart waste bins over MQTT/cellular (NB-IoT), validating message checksums, timestamps, and geolocation tags.

FR-2: 80% Fill-Level Threshold Dynamic Filtering: The system shall dynamically filter active bins, queuing for pickup only receptacles reporting a verified fill level exceeding 80% capacity, discarding sub-threshold bins from collection schedules.

FR-3: Sub-45s Fleet Route Optimization: The Route Optimization Service shall solve the multi-vehicle routing problem for 50 collection trucks across 10,000 bins in strictly under 45 seconds upon trigger, minimizing fuel consumption and travel duration.

FR-4: Real-Time In-Cab Manifest Streaming: The Driver Manifest Service shall compile turn-by-turn waypoint manifests for each truck and stream active manifests to in-cab driver applications via secure WebSockets.

FR-5: Pickup Verification & Bin Status Reset: The system shall allow drivers to confirm bin clearance in real time, immediately logging pickup completion and resetting the bin status in the spatial database.

FR-6: Citizen Missed-Pickup Complaint Management: The system shall allow citizens to log missed-pickup reports with photographic evidence and GPS coordinates, generating an auditable tracking ticket with status progression.

FR-7: Supervisor Operations Dashboard & Dispatch Override: The Web Portal shall provide supervisors with an interactive GIS map displaying live truck locations, bin fill levels (>80% highlighted in amber/red), and manual route re-dispatch controls.

4. Non-Functional Requirements

NFR-1: Performance & Computational Latency: Route recalculation for 50 trucks across 10,000 candidate bins must execute in strictly less than 45 seconds (99th percentile). Ingestion latency must not exceed 250ms per telemetry burst.

NFR-2: Scalability: The ingestion tier shall horizontally auto-scale to process over 5,000 telemetry messages per second during peak reporting intervals without packet drops.

NFR-3: Reliability & Fault Isolation: Failure in the Complaint Management Service or telemetry stream shall not impede active truck navigation or the Route Optimization Service.

NFR-4: Security & Authentication: All external API traffic shall require TLS 1.3 encryption. APIs shall enforce JWT-based RBAC with strict rate limiting to prevent denial-of-service attempts.

NFR-5: Availability: The system shall guarantee a minimum service availability of 99.9% uptime during operational sanitation shifts (05:00 - 22:00 daily).

NFR-6: Data Integrity: Spatial geometries and historical telemetry records shall adhere to ACID transaction guarantees, preventing duplicate pickup assignments.

5. Security Objectives & Traceability

- Security Objective 1 (Confidentiality & Access Control): Enforce strict boundary isolation at the API Gateway; external citizens cannot view municipal telemetry data or driver manifests.
- Security Objective 2 (Data Integrity & Authenticity): Sensor telemetry packets must include HMAC signatures to prevent spoofed fill-level reports from triggering unauthorized truck dispatches.
- Security Objective 3 (Resilience & Blast Radius Control): Database schemas are segmented per microservice with principle of least privilege, preventing lateral movement in case of service compromise.

Work Breakdown Structure (WBS)

Municipal Solid Waste Route Optimizer

Hierarchical Work Package Decomposition & Schedule | Student SRN: PES1UG24CS453

1. Executive Summary & Project Hierarchy

The Work Breakdown Structure (WBS) decomposes the development and deployment of the Municipal Solid Waste Route Optimizer into discrete, manageable deliverables and work packages across 8 key phases. Each phase defines specific milestones, deliverables, and role ownership to guarantee adherence to the sub-45s optimization SLA and municipal deployment standards.

2. Hierarchical Work Breakdown Structure (WBS Tree)

1.0 Project Management & Quality Assurance

- 1.1 Project Initiation, Scope Definition & PES SE Lab Charter
- 1.2 Agile Sprint Cadence Planning (2-week iterations, daily standups)
- 1.3 Risk Management, Quality Gates & Compliance Audits

2.0 Requirements Engineering & Analysis

- 2.1 Stakeholder Elicitation (Sanitation Supervisors, Drivers, Municipal Admin)
- 2.2 Formal IEEE 830 Software Requirements Specification (SRS)
- 2.3 Use Case Modeling, Actor Hierarchy & Requirement Traceability Matrix

3.0 System Architecture & Microservices Design

- 3.1 Architectural Pattern Evaluation & Microservices Selection
- 3.2 UML 2.0 Component Modeling (Interfaces, Ball & Socket, Connectors)
- 3.3 Polyglot Persistence Architecture (PostGIS, TimescaleDB, Redis Cache)
- 3.4 API Gateway, Service Mesh & Edge Security Specification

4.0 IoT Telemetry Ingestion & Sensor Integration

- 4.1 Ultrasonic Sensor Calibration & Payload Definition (MQTT/NB-IoT)
- 4.2 Sensor Data Ingestion Service & Telemetry Pipeline
- 4.3 Dynamic Threshold Filtering Engine (Isolate Bins >80% Full)
- 4.4 Telemetry Time-Series Persistence & Anomaly Noise Removal

5.0 Route Optimization & Algorithmic Engine

- 5.1 Urban Road Network Topology & Graph Formulation
- 5.2 Multi-Vehicle Routing Problem (VRP) Heuristic Solver
- 5.3 Distributed GPU/Compute Acceleration (Sub-45s / 50 Trucks / 10k Bins)
- 5.4 Route Request Assembly Interface (gRPC / Protobuf Endpoint)

6.0 Manifest Dispatch & User Interface Portals

- 6.1 Supervisor Web Operations Portal (GIS Fleet Tracking & Analytics)
- 6.2 Driver In-Cab Navigation Mobile App (WebSocket Manifest Sync)
- 6.3 Citizen Missed-Pickup Reporting Portal & Ticket Tracking
- 6.4 Notification & Incident Alerting Service

7.0 Verification, Validation & Testing

- 7.1 Automated Unit & Component Integration Testing
- 7.2 Rigorous 45-Second VRP Performance & Stress Benchmark Testing
- 7.3 Security Penetration Testing, JWT Validation & Role-Based Access Audit

- 7.4 Field Acceptance Testing with Sanitation Drivers & Hardware Sensors

8.0 Deployment, DevOps & Infrastructure

- 8.1 Containerization with Docker & Multi-Stage Production Builds
- 8.2 Kubernetes Cluster Orchestration & Horizontal Pod Autoscaling (HPA)
- 8.3 CI/CD Deployment Automation & GitOps Pipelines
- 8.4 Centralized Observability, Logging & Alerting (Prometheus / Grafana)

3. Work Package Dictionary & Milestone Matrix

WBS ID	Work Package	Key Deliverable	Owner Role	Duration
WP 1.0	Project Management	Agile Plan, Risk Matrix, Sprint Logs	Project Lead	2 Weeks
WP 2.0	Requirements Eng.	IEEE 830 SRS, Traceability Matrix	Systems Analyst	2 Weeks
WP 3.0	System Architecture	UML Component Diagram, Arch Justification	Chief Architect	2 Weeks
WP 4.0	IoT Telemetry Ingest	Ingestion Service, MQTT Adapter, Filter	IoT / Backend Lead	3 Weeks
WP 5.0	Route Optimization	VRP Solver, <45s Benchmark Engine, gRPC	Algorithm Specialist	4 Weeks
WP 6.0	Portals & Manifest	Supervisor Dashboard, Driver App, WS Stream	Frontend / Mobile	3 Weeks
WP 7.0	Testing & QA	Test Plan, 10k Bin Load Tests, Bug Tracker	QA / Security Lead	2 Weeks
WP 8.0	DevOps & Launch	K8s Manifests, CI/CD Pipeline, Grafana	DevOps Engineer	2 Weeks